

Module 6 — Asset Wealth & Portfolio Overview

(modules/assets)

Purpose

Capture immutable, multi-currency financial snapshots across all household assets and liabilities; compute true savings rate and net-worth trajectory; and provide per-entity attribution views. This module is the household's "net-worth scorecard," answering: *How much are we worth? Where is the money? How fast are we building wealth? What portion is earned vs. market-driven?*

Actors & User Stories

- *As a household head*, I review my net-worth dashboard monthly; see allocation across accounts/investments/real estate; and identify where savings are growing fastest.
- *As an investor*, I link my brokerage holdings (units × unit price) and see real-estate valuations update alongside liquid accounts; I distinguish contribution from market-driven gains.
- *As a planner*, I run FIRE/coast-FIRE projections: given my current savings rate, how long until financial independence?
- *As a reviewer*, I validate annual snapshots (pulling live account balances) and correct outliers due to temporary market swings.
- *As a privacy advocate*, I control whether optional price refresh (stocks, crypto, forex) is enabled; it defaults **off**.

Data Model

All monetary fields (`_eur` suffix) are **NUMERIC(10,2)** decimal EUR amounts (e.g. **19.99**), never minor-unit integers; UI displays them pt-PT-formatted (e.g. **19,99€**).

- **Asset**: `id`, `entity_id`, `name`, `asset_class`(CHECKING|SAVINGS|INVESTMENT|REAL_ESTATE|VEHICLE|CRYPTO|FOREX|LIABILITY), `currency`, `iban_masked?`, `ticker?`, `description`, `is_deleted`, `linked_account_id?`.
- **AssetSnapshot** (immutable, append-only): `id`, `asset_id`, `as_of_date`, `value_eur`, `fx_rate_to_eur` NUMERIC(10,6), `source`(MANUAL|DERIVED|LINKED_ACCOUNT_AUTO_SYNC), `snapshot_batch_id`, `created_at`. Every correction is a new row.
- **CashFlowPeriod** (derived from M2): `id`, `entity_id`, `period_start`, `period_end`, `income_eur`, `expense_eur`, `net_savings_eur`, `savings_rate` NUMERIC(4,3). Savings rate = $(income - expense) / income$.
- **FxRate** (historical, immutable): `id`, `from_currency`, `to_currency`, `rate` NUMERIC(10,6), `as_of_date`, `source`(MANUAL|PROVIDER|HISTORICAL_ECB).
- **Holding** (investments): `id`, `asset_id`, `ticker`, `quantity` NUMERIC(14,8), `unit_value_eur` NUMERIC(18,6) (finer precision for crypto/forex sub-cent prices), `last_updated_at`. Price refresh optional, off by default.
- **Liability** (or `asset_class=LIABILITY`): `id`, `entity_id`, `name`, `principal_eur`, `interest_rate_annual`, `term_months`, `issue_date`, `currency`,

`creditor_merchant_id, status(ACTIVE|PAID_OFF), description.`

- **LiabilityAmortization** (derived, append-only): `id, liability_id, period_start, period_end, principal_paid_eur, interest_paid_eur, remaining_balance_eur.`
- **SnapshotBatch**: `id, entity_id, period, status, executed_at, note.`

Functional Requirements

- **FR-6.1 Periodic Snapshot Scheduling.** Celery Beat (configurable cadence) fetches balances from linked accounts (M2) and creates AssetSnapshot rows. Non-linked assets manual or bulk import. High-confidence auto-accepted; manual → review.
- **FR-6.2 Inflow/Outflow & True Savings Rate.** Consume **Transaction** ledger per entity/period; compute income (credit tagged income/salary/bonus/transfer_in), expense (rest), net savings, savings rate; handle zero/negative income. Store in **CashFlowPeriod**.
- **FR-6.3 Net-Worth Analytics.** Aggregate active assets by entity/date: net-worth = $\Sigma(\text{asset snapshots}) - \Sigma(\text{liability balances})$. Time-series (1yr/5yr/all-time); allocation pie by class; entity filters.
- **FR-6.4 Multi-Currency Consolidation.** Foreign snapshots store `fx_rate_to_eur` at snapshot date; EUR default with original-currency drill-down.
- **FR-6.5 Investment Holding Valuation.** **Holding** links asset↔ticker; `value_eur = quantity × unit_value_eur`. Optional price refresh (opt-in) is logged, **never** auto-updates holdings.
- **FR-6.6 Liability Principal & Interest Decomposition.** Auto-compute amortization; monthly snapshots record principal vs interest; display liabilities as negative net worth; paydown as separate trend.
- **FR-6.7 Real-Estate Manual Valuation.** REAL_ESTATE assets allow periodic manual valuation; track history; surface cadastral reference + last appraisal date.
- **FR-6.8 Crypto & Forex Spot Holdings.** CRYPTO/FOREX store quantity + spot value; optional refresh if enabled. Privacy default off.
- **FR-6.9 Correction & Audit Trail.** Snapshots immutable; corrections spawn new `source=MANUAL` snapshot + reason; AuditLog records each. UI shows history + reasons.
- **FR-6.10 Savings Goal Tracking.** Optional target net-worth + timeline; dashboard overlays progress; projects forward on current rate.
- **FR-6.11 FIRE / Coast-FIRE Projections.** From net-worth, savings rate, assumed real return (default 5%), compute years to FI ($25 \times \text{spend or target}$); coast scenario supported.
- **FR-6.12 Account-to-Asset Linking.** Link **BankAccount** (M2) to **Asset** so snapshots auto-derive; idempotent; no future dating.
- **FR-6.13 Contribution vs. Market Attribution.** Store `contributed_eur`; `market_gain_eur = value - contributed`; dashboard shows earned vs market growth.

Automation Rules

1. **Scheduled Snapshots** (Celery Beat): daily 02:00 UTC for linked accounts; manual for others; on error → ProcessingJob + alert.
2. **FX Rate Refresh**: daily 09:00 UTC; 6-month historical for EUR/USD, EUR/GBP, EUR/CHF, EUR/JPY (ECB or open source).
3. **Amortization Auto-Compute**: on liability create/edit, compute term schedule; monthly snapshots derive balance.
4. **Cash Flow Derivation**: daily recompute of prior-month **CashFlowPeriod** (idempotent) from confirmed M2 transactions.
5. **Anomaly Alert**: net-worth drops >15% in a month without explainable spike → review.

6. **Price Refresh (Opt-In, Off by Default):** weekly task fetches latest prices for INVESTMENT/CRYPTO; results **logged only**, never auto-update; non-blocking on failure.

UI / Screens

1. **Net-Worth Dashboard:** line chart (YTD + all-time); current total; YoY %; allocation pie; savings rate, contribution vs gain, months to FI.
2. **Asset Manager:** list w/ latest value; add/edit/delete; link account; price-refresh opt-in; real-estate appraisal upload.
3. **Snapshot History:** timeline per asset; prior values + edit reasons; manual entry.
4. **Allocation Donut:** drill-down; compare vs target allocation.
5. **Savings Rate Trend:** 12-month line; target overlay; highlight below-target months.
6. **Liability Dashboard:** remaining balance, interest paid YTD, payoff date, amortization detail.
7. **Projection Panel:** FIRE timeline, coast-FIRE scenarios; sliders for return and retirement spend.
8. **Entity Selector:** any single entity, or "todas" (global, persistent).

API Surface

POST/GET/PATCH /assets · GET/POST /assets/{id}/snapshots · GET /assets/net-worth?entity_id=&as_of_date= · GET /cashflow?entity_id=&period= · POST /assets/{id}/correct-snapshot · CRUD /liabilities · GET /liabilities/{id}/amortization · CRUD /holdings · POST /holdings/{id}/refresh-price (opt-in, non-blocking) · GET /analytics/projection.

Analytics & KPIs

- Net-worth over time (multi-year line).
- Asset allocation (pie by class; donut by account).
- True savings rate $((\text{income} - \text{expense}) / \text{income})$, monthly + YTD).
- Contribution vs market gain (stacked bar).
- Liability paydown (waterfall: principal/interest/remaining).
- YoY net-worth growth (% , CAGR).
- Months to FIRE.
- Real-estate appreciation (€ and % per property).
- Cash-flow sustainability (months of expenses in liquid reserves).

Edge Cases & Validation

1. **Snapshot immutability:** only latest per (asset, date) corrected; priors read-only; corrections spawn rows + reason.
2. **Liabilities reduce net worth:** $\Sigma(\text{assets}) - \Sigma(\text{liability.remaining_balance})$.
3. **Multi-currency:** each snapshot stores `fx_rate_to_eur`; EUR sum + optional original-currency display.
4. **Linked account idempotency:** as-of-date check prevents duplicates; balance change → new snapshot.
5. **Negative snapshots:** overdrafts valid; treated as liabilities.
6. **Zero/negative income:** savings rate null / N/A; no division by zero.
7. **Future dates rejected:** snapshots ≤ today.
8. **Stale price data:** >7 days old → UI warning; value not silently degraded.

9. **Correction reasoning:** every manual snapshot has a **note**; audit surfaces it.

Additional / Enriched Requirements

- **Investment Holdings with Quantity Valuation:** security-granular tracking without live feeds; optional refresh non-destructive (logged only), off by default.
- **Liability Amortization & Interest Attribution:** term/rate → amortization tables; principal vs interest split; "true cost of debt".
- **Real-Estate Valuation Audit Trail:** appraisal dates, cadastral references, manual valuations; multi-property.
- **Crypto & Forex Micro-Holdings:** quantity + unit price; privacy-respecting optional feeds.
- **Savings Goal & FIRE Projections:** target net-worth + retirement date; interactive scenarios (5%/7%/10% real return).
- **Contribution Attribution:** cumulative contributions per asset; market gain = value – contributed.
- **Entity-Scoped Net-Worth:** filter net worth to a single entity or view the whole household; filtering at query time, not a permission check.
- **Real Return Assumption (Configurable):** default 5%; sensitivity analysis.
- **Annual Snapshot Validation Ritual:** yearly review prompt to confirm/correct holdings.

Open Questions / Decisions

1. **Price refresh provider?** → ECB (forex) + Alpha Vantage free tier default; premium optional.
2. **FIRE target formula?** → Default 25× annual spend, override option.
3. **Crypto holdings tracking?** → Manual entry only (privacy); optional Koinly-style import if opt-in.
4. **Liability interest compounding?** → Monthly.
5. **Snapshot confidence threshold?** → First 3 months manual → review; then auto-accept if consistent.
6. **Forex rounding?** → Store all EUR amounts as **NUMERIC(10,2)** (never minor-unit integers); original-currency amount stored alongside **fx_rate_to_eur** for drill-down, presented with currency-specific decimals.

Definition of Done

- ☐ Asset + Liability CRUD (all fields).
- ☐ AssetSnapshot immutability; correction spawns new rows.
- ☐ Scheduled snapshot engine (Celery Beat) daily linked accounts.
- ☐ FxRate populated daily (ECB, 6-month history); snapshot lookup works.
- ☐ CashFlowPeriod nightly from M2; savings-rate formula unit-tested.
- ☐ Net-worth aggregation: ≥10 entities × 5 years in <200ms.
- ☐ Liability amortization auto-computed; monthly snapshots match (±0.01€).
- ☐ Holding valuation tested.
- ☐ FIRE/coast-FIRE logic unit-tested.
- ☐ UI dashboard renders (net-worth line, allocation pie, savings trend, FIRE timeline).
- ☐ Price refresh (opt-in, off default) logs without modifying holdings.
- ☐ Entity filtering tested (single entity vs. whole household totals).
- ☐ Audit log for every snapshot create/correct.
- ☐ E2E: create asset → 3 snapshots → correct latest → verify history + audit.

Integration Contract

- **Exposes:** net-worth series ([GET /assets/net-worth](#)) + cash-flow ([GET /cashflow](#)) for Dashboards; [Asset/AssetSnapshot](#) as link targets for Banking (M2); liability balances as negative contributors.
- **Consumes:** [Transaction](#) cash-flow (M2), [BankAccount](#) (M2) for auto-sync, [Entity](#) (household scope), [FxRate](#), [Merchant](#) (creditors), and [Document](#) (appraisal attachments).